INT396 · UNSUPERVISED LEARNING

UNIT 1

Foundations of Unsupervised Learning

What you can learn from data when nobody has told you the answer - and why the whole thing rests on how you choose to measure 'similar'.

Supervised vs Unsupervised vs Semi-Supervised • Problem Formulation • Market Segmentation • Anomaly & Fraud Detection • Euclidean • Manhattan • Cosine • Why Metric Choice Matters

UNIT I ROADMAP

What This Unit Covers

UNIT 1

Four topics. The last one is the foundation the rest of the course stands on.

- Learning paradigms
  - Supervised vs. Unsupervised vs. Semi-Supervised learning
  - What each needs, what each produces, and when each applies
- Problem formulation
  - How to state an unsupervised problem precisely
  - Why there is no ground truth - and what you do about it
- Real-life use cases
  - Market segmentation and customer behaviour analysis
  - Anomaly and fraud detection
  - Pattern discovery in biological and social data
- Distance and similarity metrics
  - Euclidean, Manhattan and Cosine similarity
  - When and WHY the choice of distance changes your answer

1 · Three paradigms
what labels do you have?

2 · Problem formulation
five things you must state

3 · Real use cases
segmentation, fraud, biology

4 · Distance metrics
the foundation under all of it

each one depends on the one below it

SHOW OF HANDS Which of these four have you met before today?

LEARNING OUTCOMES

Learning Outcomes

UNIT 1

By the end of this unit you should be able to:

1. **Distinguish** supervised, unsupervised and semi-supervised learning, with an example of each.
2. **Formulate** an unsupervised problem using all five specifications.
3. **Justify** an evaluation strategy given that no ground truth exists.
4. **Compute** Euclidean, Manhattan and cosine distance by hand for a given pair.
5. **Select** and defend a distance metric and a scaling decision for a new dataset.

Remember
recall the terms

Understand
explain them in your own words

Apply
use them on a problem you have not seen

Analyse
choose between them and justify it

the levels this unit is assessed at

01

SECTION

Supervised vs Unsupervised vs Semi-Supervised

IN THIS SECTION

- Learning with a teacher, and learning by self-exploration
- What a 'label' actually is, and why labels are expensive
- The three paradigms side by side
- Where reinforcement learning fits
- Classifying unfamiliar problems

Supervised Learning - Learning With a Teacher

You already know the answer for the training data. The job is to generalise it.

What it needs and does

Input features X together with a LABEL y for every single example.

The task learn a mapping f such that $f(X) \approx y$, then apply it to data you have never seen.

Two families CLASSIFICATION predicts a category; REGRESSION predicts a number.

Success is measurable hold back some labelled data, predict it, compare with the truth. Accuracy, precision, recall.

The catch somebody had to produce every one of those labels, by hand.

The spam filter, concretely

The data 10,000 emails. Each one already stamped SPAM or NOT SPAM by a human being.

The features X words present, number of links, sender reputation, time sent.

The label y SPAM / NOT SPAM. Binary. Unambiguous. Given.

What is learned a decision rule that maps a new, unseen email to one of the two classes.

Why it works the categories were decided BEFORE learning started. The algorithm only had to find the boundary.

Features X what you measure → **+ Label y** supplied by a human → **Learn f** so that $f(X) \approx y$ → **Predict** on unseen data

NAME ONE a supervised task you personally used today - and say who produced its labels.

Unsupervised Learning - Learning by Self-Exploration

UNIT 1

No labels. No teacher. No answer key - and therefore no single correct answer.

What it needs and does

Input features X only. There is NO y. Nothing is labelled.

The task discover structure that is already present in the data - groups, patterns, outliers, or a lower-dimensional shape.

Main families CLUSTERING (group similar items), DIMENSIONALITY REDUCTION (compress features), ASSOCIATION (find co-occurrence), ANOMALY DETECTION (find the odd ones).

Success is hard to measure there is no truth to compare against. You need internal measures, or a human, or a downstream business result.

The pay-off it works on data nobody has labelled - which is almost all data that exists.

The library, concretely

The situation a room with 5,000 books dumped on the floor. No Dewey system. No genre stickers. No catalogue.

The task organise them so the room is usable.

What you would do flick through, notice some are about cooking, some about history, some are children's picture books - and start piles.

The key point nobody told you 'cooking' was a category. You INFERRED the categories from the books themselves.

The uncomfortable part your colleague sorts them by size and colour instead - and produces a beautiful, usable room. Who is wrong?

Features X only — no label column → **Find structure** — groups, shapes, outliers → **You name them** — meaning is YOUR job → **No accuracy score** — nothing to compare against

THINK-PAIR-SHARE 60 seconds: name a dataset you personally generate every day that has NO labels attached to it.

1.1 LEARNING PARADIGMS

Semi-Supervised Learning - A Few Labels, and a Lot of Data

UNIT 1

The realistic middle ground - and in industry, the most common situation of the three.

What it is and why it exists

Input a SMALL amount of labelled data plus a LARGE amount of unlabelled data.

Why it exists labels are expensive and slow; raw data is cheap and abundant. This is the normal situation, not the exception.

Core assumption points close together in feature space probably share a label. The unlabelled data reveals the SHAPE; the few labels tell you what to call each region.

Typical methods self-training (label confidently, retrain), and label propagation across a graph of similar points.

The risk one wrong confident label can propagate through the whole dataset and poison it.

Medical imaging, concretely

The data 50,000 chest scans in a hospital archive. All unlabelled.

The labels available 300 of them have been read and annotated by a consultant radiologist.

Why not just label them all at ~10 minutes per scan, 50,000 scans is over 8,000 consultant-hours. Nobody is paying for that.

What semi-supervised does uses all 50,000 to learn what scans generally LOOK like, and the 300 to attach clinical meaning to the structure it found.

The result far better than 300 labelled scans alone, at a tiny fraction of the labelling cost.

UNLABELLED
93%
50,000 scans - gives the SHAPE

LABELLED
7%
300 - gives the MEANING

structure from the many, meaning from the few

DO THE SUM At ~10 minutes a scan, how long would labelling all 50,000 take one radiologist?

1.1 LEARNING PARADIGMS

The Three Paradigms Side by Side

UNIT 1

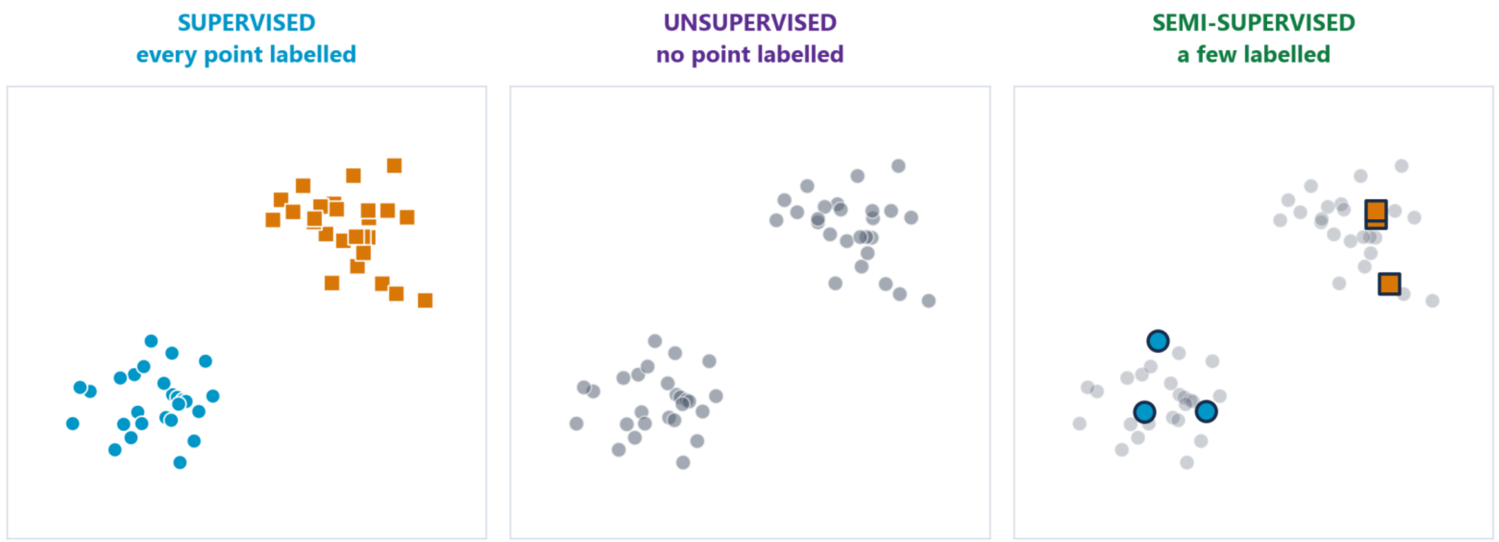
Same data, three different situations - decided entirely by which labels you happen to have.

	Supervised	Unsupervised	Semi-Supervised
Labels needed	Every example	None at all	A small fraction
The goal	Predict a known target	Discover unknown structure	Predict, using structure to stretch few labels
Typical output	A class or a number	Groups, components, outliers	A class or a number
Evaluation	Direct: accuracy vs. truth	Indirect: internal measures or human judgement	Direct, on the labelled held-out part
Cost driver	Human labelling effort	Compute and interpretation	A little labelling, a lot of data
Example	Spam filter from labelled emails	Grouping customers with no predefined segments	Diagnosing scans from 300 annotated out of 50,000

ONE QUESTION Look at your own project data. Which row are you in - and how do you know?

UNIT 1

The Three Paradigms, Side by Side



HOW TO READ THIS FIGURE

Left - supervised every point carries its label. Shape and colour = the class you were told.

Middle - unsupervised identical points, all grey. Nobody told you anything.

Right - semi-supervised a handful of labelled points against a crowd of unlabelled ones.

The data is the same in all three panels. Only the LABELS differ.

Which is why the paradigm is decided by what labels you happen to have - not by the algorithm you prefer.

TAKEAWAY Same points, three panels. The paradigm is a property of your labels, not of your data.

A Note on the Fourth Paradigm: Reinforcement Learning

UNIT 1

EXAM TIP · Not on this syllabus - but you should know why it is different

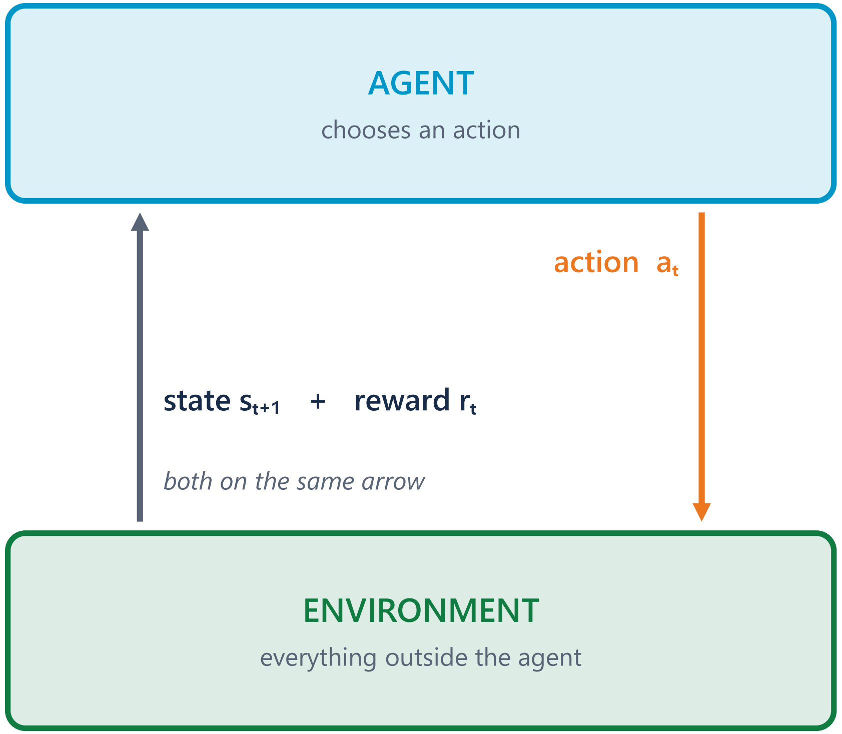
What it is an agent takes actions in an environment and receives REWARDS or penalties. It learns a policy that maximises reward over time.

Why it is not unsupervised there IS a signal - the reward. It is just delayed, sparse and evaluative rather than a per-example label.

Why it is not supervised nobody tells you the correct action. You are told how good the outcome was, AFTER the fact, sometimes much later.

The one-line distinction supervised = 'the answer was cat'. reinforcement = 'that went badly'. unsupervised = silence.

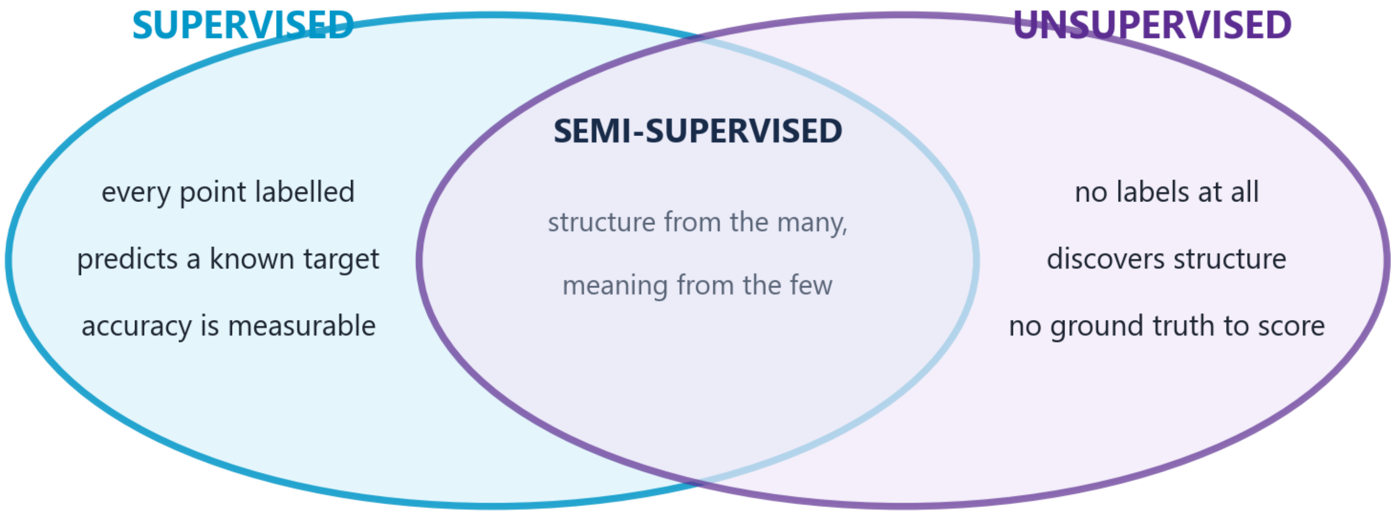
Where you will meet it game playing, robotics, recommendation, resource scheduling. Not in this course - but do not confuse it with unsupervised learning in an exam.



the reward comes FROM the environment

SORT THEM 'the answer was cat' / 'that went badly' / silence - which is which?

What Each Paradigm Needs, as a Venn Diagram



HOW TO READ THIS FIGURE

The overlap is the point semi-supervised is not a third thing - it is genuinely both at once.

From the unsupervised side it uses all the unlabelled data to learn the SHAPE.

From the supervised side it uses the few labels to attach MEANING to that shape.

Core assumption points close together in feature space probably share a label.

Why it matters in industry this overlap is the normal situation, not the exception.

TAKEAWAY **Structure from the many, meaning from the few. That division of labour is what semi-supervised buys you.**

Which Paradigm? Four Real Problems

THE PROBLEM

- A bank wants to know which loan applicants will default, from 40,000 past applications with known outcomes
- An e-commerce site wants to find natural customer groups - it has no idea how many groups exist
- A hospital has 50,000 scans, of which 300 are annotated
- A telecom company wants to find SIM cards behaving unlike any other SIM card on the network

WHY IT MATTERS

- 'We don't know how many groups' is always an unsupervised signal
- 'Find the unusual ones' with no list of known-bad = anomaly detection
- A few labels against a lot of data = semi-supervised, every time

THE TELL

S	U	SS	?
labels present and complete	no target column at all	a few labels, lots of raw data	always look at the spreadsheet first

STEP-BY-STEP SOLUTION

Loan default SUPERVISED - classification. Every past application has a known outcome, which is the label y. Target is defined and filled in.

Customer groups UNSUPERVISED - clustering. There is no 'correct' segment column anywhere. Note the giveaway phrase: 'no idea how many groups exist'.

Hospital scans SEMI-SUPERVISED. A few hundred labels against fifty thousand unlabelled examples - the classic ratio.

Odd SIM cards UNSUPERVISED - anomaly detection. Nobody has a list of which SIMs are fraudulent; if they did, it would be supervised.

The tell in every case look for the target column. Present and complete → supervised. Absent → unsupervised. Sparse → semi-supervised.

ANSWER **The paradigm is decided by the labels you have, not by the algorithm you prefer.**

CHECK YOUR UNDERSTANDING

Quick Check - Section 1

UNIT 1

QUICK CHECK

A streaming service groups its users into 'listening tribes'. Nobody defined the tribes in advance, and nobody knows how many there should be. Which paradigm is this?

- **A** Supervised learning - the listening history is the label
- **B** Unsupervised learning - clustering
- **C** Semi-supervised learning - some users have stated preferences
- **D** Reinforcement learning - the service is rewarded by watch time

ANSWER: B There is no target column. The tribes do not exist until the algorithm creates them, and even the NUMBER of tribes is unknown. That is clustering, which is unsupervised. Listening history is a FEATURE (X), not a label (y) - confusing the two is the most common error in this topic.

FEATURE (X)	LABEL (y)
listening history	does not exist
what you MEASURE	what you PREDICT

confusing these two is the classic error

02

SECTION

Problem Formulation for Unsupervised Learning

IN THIS SECTION

- Why formulation is HARDER without labels
- The five things you must specify before you cluster
- A worked formulation: e-commerce segmentation
- The evaluation problem, and three honest answers to it

Why Formulation Is Harder Without Labels

COMMON PITFALL · The label was doing more work than you realised

In supervised learning, the label pins everything down it defines what the task is, what counts as success, which features matter, and when you are finished.

Remove the label and all four of those become YOUR decisions and every one of them is a judgement call you must be able to defend.

Decision 1 - What does 'similar' mean? the algorithm has no opinion. You supply it by choosing features and a distance metric. This is section four.

Decision 2 - How many groups? there is rarely a correct answer. 3 segments and 30 segments can both be defensible for the same customers.

Decision 3 - What counts as a good result? no accuracy score exists. You need internal measures, human judgement, or a downstream outcome.

Decision 4 - When do you stop? without a target metric, 'good enough' has to be defined by the business question, not by the data.

THE LABEL USED TO ANSWER ALL FOUR

1. What does 'similar' mean?
2. How many groups?
3. What counts as a good result?
4. When do you stop?

THINK ABOUT IT If two teams cluster the same customers and get completely different groups - is one of them wrong?

Formulating an Unsupervised Problem: Five Specifications

Five things you must state explicitly, because nothing in the data will state them for you.

SPECIFICATION 1

The objects

- WHAT are you grouping or analysing?
- Customers? Transactions? Documents? Genes?
- One row of your data = one object
- Getting this wrong invalidates everything after it

SPECIFICATION 2

The features

- WHICH measurable properties describe each object?
- Age, income, visits/month, average basket value
- Every feature you include shapes what 'similar' means
- Irrelevant features actively add noise

SPECIFICATION 3

The similarity measure

- HOW do you decide two objects are alike?
- Euclidean, Manhattan, Cosine - section four
- Includes the scaling decision, which is not optional
- The single most consequential choice you make

SPECIFICATION 4

The structure sought

- WHAT KIND of pattern are you looking for?
- Groups? A lower-dimensional shape? Outliers? Rules?
- This chooses the family of algorithm
- Clustering vs. dimensionality reduction vs. anomaly detection

SPECIFICATION 5

The success criterion

- HOW will you judge the result - before you run it?
- Internal measure, human review, or business outcome
- Decide this FIRST, or you will rationalise whatever you get
- The hardest specification, and the most skipped

TOGETHER

= A well-posed problem

- Objects · Features · Similarity · Structure · Success
- Write all five down BEFORE opening your editor
- Any one left implicit will be decided by accident
- This is your checklist for every practical this semester

APPLY IT Pick any dataset you know. Say its OBJECT out loud in three words.

Worked Formulation: E-Commerce Customer Segmentation

THE PROBLEM

- An online retailer has 2 years of order history
- Marketing wants to 'understand our customers better'
- There are no existing segments and no target column
- Task: turn that vague request into a well-posed problem

WHY IT MATTERS

- 'Understand our customers better' is not a problem statement
- Aggregate to the right OBJECT before anything else
- A success criterion agreed in advance stops you rationalising the output

THE CLASSIC STARTING POINT

R	F	M
Recency - days since last order	Frequency - orders per year	Monetary - average basket

STEP-BY-STEP SOLUTION

1. Objects the CUSTOMER, not the order. Aggregate the order table to one row per customer first - this step is easy to skip and fatal.

2. Features recency (days since last order), frequency (orders per year), monetary (average basket value), category spread, discount sensitivity. The classic RFM starting point.

3. Similarity Euclidean distance ON STANDARDISED features - because monetary value is in thousands of rupees and frequency is in single digits.

4. Structure sought clusters. Try k from 2 to 8 and inspect - the business can realistically act on 3 to 6 segments.

5. Success criterion agreed in advance: each segment must be describable in one sentence a marketer understands, and must differ measurably in at least two features.

Why criterion 5 is the good one it is a BUSINESS test, not a mathematical one. A statistically excellent clustering nobody can act on has failed.

ANSWER Vague request → five explicit specifications → a problem you can actually solve.

The Evaluation Problem - How Do You Know It Worked?

UNIT 1

No ground truth does not mean no evaluation - it means you must choose your evidence deliberately.

Why this is genuinely hard

The core difficulty there is no ground truth. Nobody wrote down the correct answer, so there is nothing to compare against.

You cannot report accuracy '94% accurate' is meaningless when no correct grouping exists.

Different valid answers changing the metric, the features or k gives a different clustering - and several may be defensible.

The temptation run it, look at the output, then invent a story about why it is good. This is extremely easy to do accidentally.

Three honest ways to evaluate

1. Internal measures score the geometry itself - are clusters tight inside and far apart outside? Silhouette score, within-cluster sum of squares. No labels required.

2. External / human judgement show the segments to a domain expert. Can they name each one? Do they recognise them? Unglamorous and very effective.

3. Downstream outcome the strongest test. Run a campaign targeted by segment and measure whether response improved. The clustering is good if the DECISION it enabled was good.

Internal
is the geometry tight?
silhouette

→

Human
can an expert
name the groups?

→

Downstream
did the DECISION
improve?

RANK THEM Which of the three would convince YOU? Which would convince a manager who does not know what a silhouette score is?

CHECK YOUR UNDERSTANDING

Quick Check - Section 2

UNIT 1

QUICK CHECK

You cluster customers and get a silhouette score of 0.71 - geometrically excellent. Marketing looks at the segments and cannot describe a single one of them. What is the correct conclusion?

A The clustering is good; marketing needs training

B The clustering is bad; the silhouette score must be wrong

C The clustering is geometrically sound but has failed its success criterion

D You should increase k until the segments become describable

ANSWER: C A high silhouette score says the clusters are tight and well separated - a statement about GEOMETRY only. It says nothing about meaning. If the agreed success criterion was 'describable and actionable', the result has failed that criterion, and the honest next step is to revisit the FEATURES, not to blame the audience.

GEOMETRY

0.71

silhouette says: excellent

MEANING

none

marketing can name nothing

both statements are true at the same time

03

SECTION

Real-Life Use Cases

IN THIS SECTION

- Market segmentation
- Customer behaviour analysis
- Anomaly and fraud detection
- Pattern discovery in biological and social data
- What all four have in common

Market Segmentation and Customer Behaviour Analysis

USE CASE 1

Market segmentation

- Divide a customer base into groups with similar behaviour
- Input: RFM, demographics, category mix, channel
- Output: 3-6 named, actionable segments
- Value: one message per segment beats one message for everyone

USE CASE 1

What it looks like in practice

- 'Bargain hunters' - only ever buy during sales
- 'Loyal regulars' - small, frequent, full-price purchases
- 'Big occasional' - twice a year, very large baskets
- 'Lapsed' - were frequent, nothing for 6 months

USE CASE 2

Customer behaviour analysis

- Discover PATTERNS in what people do, not who they are
- Sequence of pages before a purchase; time-of-day patterns
- Association rules: what gets bought together
- Basket analysis is the classic example

USE CASE 2

Why it is unsupervised

- Nobody wrote down the list of behaviours to look for
- The patterns are discovered, not verified
- Often surprising - which is the whole point
- A supervised model can only confirm what you suspected

WHAT GOOD SEGMENTS LOOK LIKE

- **B** bargain hunters - only buy in a sale
- **L** loyal regulars - small, frequent, full price
- **O** big occasional - twice a year, huge basket
- **X** lapsed - were frequent, now nothing

BRAINSTORM Name a segment YOU belong to as a customer of some app. What behaviour defines it?

Anomaly Detection, Fraud, and Pattern Discovery

UNIT 1

Two use cases where unsupervised learning is not just convenient - it is the only option.

USE CASE 3

Anomaly and fraud detection

- Find the points that do not resemble anything else
- Card fraud, network intrusion, machine failure, insurance abuse
- Anomalies are RARE - often well under 1% of records
- Which is exactly why labels are scarce

USE CASE 3

Why unsupervised beats supervised here

- A supervised model learns PAST fraud patterns
- So it catches fraud that resembles fraud you already know
- New schemes look like nothing in the training set
- Unsupervised catches 'unlike anything' - including the new

USE CASE 4

Biological data

- Cluster gene expression profiles across patients
- Groups emerge that correspond to DISEASE SUBTYPES
- Nobody knew the subtypes existed beforehand
- Different subtypes respond to different treatments

USE CASE 4

Social data

- Community detection in social networks
- Topic discovery across millions of documents
- Detecting coordinated inauthentic behaviour
- Structure in who-connects-to-whom, not in what they say

FOUR PLACES LABELS DO NOT EXIST

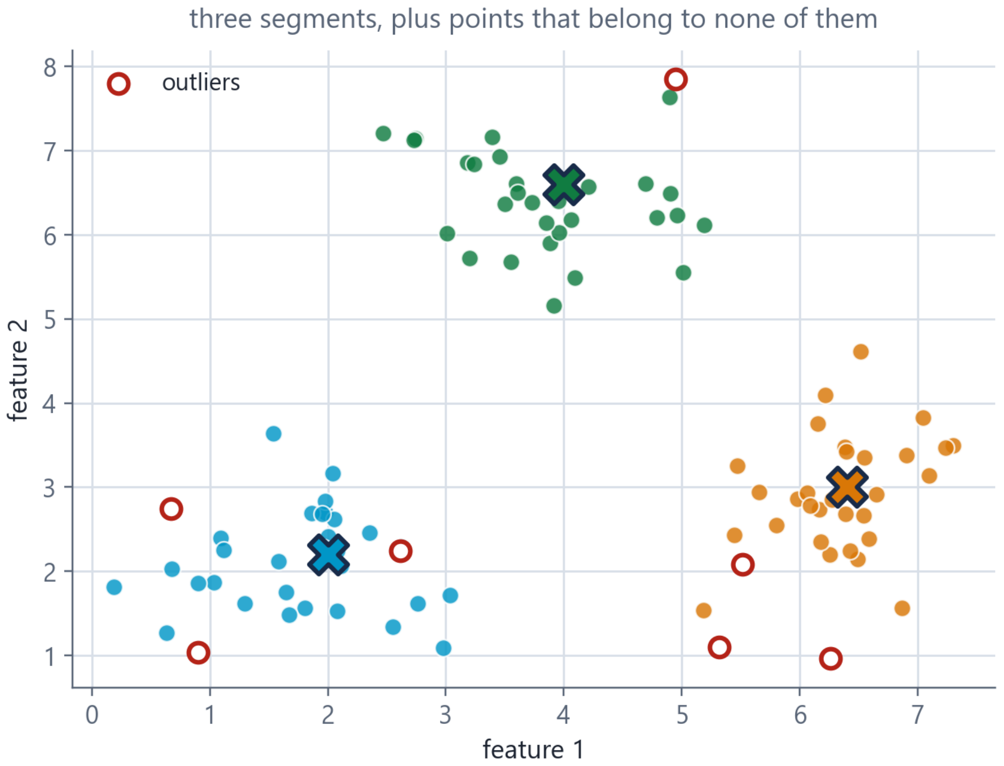
! fraud - unlike anything seen before

~ faults - nobody labels 'about to fail'

DNA disease subtypes nobody knew existed

@ communities in a social graph

What Clustering Actually Produces



HOW TO READ THIS FIGURE

The coloured groups the segments. Nobody defined them - the algorithm found them.

The X marks each cluster's centre. In K-Means this is literally the mean of its members.

The hollow red circles points far from every group - candidate anomalies.

Two use cases, one picture segmentation reads the clusters; anomaly detection reads the leftovers.

The catch the axes are unlabelled here. On real data YOU chose those features - and that choice made these groups.

TAKEAWAY **Segmentation and anomaly detection are the same computation read two different ways.**

Case Study: Detecting Fraudulent Card Transactions

THE PROBLEM

- A bank processes millions of card transactions per day
- Confirmed fraud is roughly 0.1% of transactions
- Confirmed labels arrive weeks late, via customer complaints
- Task: design an unsupervised approach

WHY IT MATTERS

- Rare-event problems make labels scarce AND late - a double bind
- Defining 'normal' per-entity is usually stronger than a global rule
- The false-alarm cost is a business decision, not a technical one

EVERY FEATURE IS RELATIVE

1	2	3	4
amount vs THIS customer's normal	distance from THEIR usual places	hour vs THEIR pattern	merchant category novelty

STEP-BY-STEP SOLUTION

1. Objects the TRANSACTION - but described relative to that cardholder's own history, not against the population.

2. Features amount vs. that customer's usual amount; distance from their usual locations; time of day vs. their pattern; merchant category novelty; time since previous transaction.

3. Similarity distance in a standardised feature space - amounts and distances live on wildly different scales.

4. Structure sought ANOMALIES, not clusters. Points far from every dense region of that customer's normal behaviour.

5. Success criterion agreed in advance: catch a useful share of confirmed fraud while keeping false alarms low enough that genuine customers are not blocked.

The key design insight 'normal' is defined PER CUSTOMER. Rs. 40,000 is unremarkable for one cardholder and a screaming alarm for another. A global threshold fails both.

ANSWER Anomaly detection with no labels — because the labels arrive weeks too late to be useful.

What All Four Use Cases Have in Common

KEY CONCEPT · The shape of a genuine unsupervised problem

1. The categories do not exist yet segments, behaviour patterns, anomaly types and disease subtypes are all OUTPUTS, not inputs. Nobody could have supplied them.

2. Labelling is impossible, impractical or too slow you cannot label 'which segment' because segments do not exist; fraud labels arrive weeks late; nobody knew the disease subtypes.

3. There is a lot of unlabelled data and it is cheap. This is the resource unsupervised learning is designed to exploit.

4. Discovery has real value finding something you did not think to ask about is the point. A supervised model can only confirm what you already suspected.

5. Success is judged downstream by the campaign response, the fraud prevented, the treatment chosen - not by an accuracy score.

Categories do not exist yet
they are the OUTPUT

Labelling is impractical
or arrives far too late

Unlabelled data is abundant
and cheap

Discovery has value
you learn what you did not ask

Judged downstream
by the decision it enabled

APPLY IT Think of a problem from your own life or work. Does it have all five? If not, which one is missing - and is unsupervised learning really the right tool?

04

SECTION

Distance and Similarity Metrics

IN THIS SECTION

- Why the metric IS the algorithm's definition of 'similar'
- Euclidean distance - the crow flies
- Manhattan distance - the city blocks
- Cosine similarity - direction, not magnitude
- Hands-on calculation on all three

The Distance Metric IS the Definition of 'Similar'

KEY CONCEPT · **The algorithm has no opinion. You supply one.**

What a distance metric does takes two objects, each described by a list of numbers, and returns a single number: how far apart they are.

Why it is not a detail clustering means grouping things that are close. Change what 'close' means and the groups change - same data, same algorithm, different answer.

The four properties of a true metric non-negativity $d(a,b) \geq 0$; identity $d(a,a) = 0$; symmetry $d(a,b) = d(b,a)$; and the triangle inequality $d(a,c) \leq d(a,b) + d(b,c)$.

Similarity vs. distance they are inverses. High similarity = small distance. Cosine gives SIMILARITY (higher is more alike); Euclidean and Manhattan give DISTANCE (lower is more alike). Do not mix them up.

The practical consequence before you cluster anything, you must be able to finish this sentence: 'two of my objects are similar when ___'.

A TRUE METRIC OBEYS FOUR RULES

1. non-negativity $d(a,b) \geq 0$
2. identity $d(a,a) = 0$
3. symmetry $d(a,b) = d(b,a)$
4. triangle $d(a,c) \leq d(a,b) + d(b,c)$

FINISH THE SENTENCE 'Two of my objects are similar when ___.' Say yours out loud.

Euclidean Distance - How the Crow Flies

The default metric - and the right one when your features are continuous and comparably scaled.

EUCLIDEAN DISTANCE IN 2 DIMENSIONS

$$d(p, q) = \sqrt{[\,(p_1 - q_1)^2 + (p_2 - q_2)^2\,]}$$

Pythagoras. The straight-line distance between two points.

EUCLIDEAN DISTANCE IN N DIMENSIONS

$$d(p, q) = \sqrt{[\,\Sigma_{i=1}^{n} (p_i - q_i)^2\,]}$$

Same idea, more features. Also written as the L2 norm $\|p - q\|_2$.

WORKED: P = (2, 3), Q = (6, 6)

$$d = \sqrt{[\,(2-6)^2 + (3-6)^2\,]} = \sqrt{[16 + 9]} = \sqrt{25} = 5$$

The classic 3-4-5 triangle. Do this one by hand now.

WHERE, AND WHEN TO USE IT

p, q two objects, each a list of n feature values

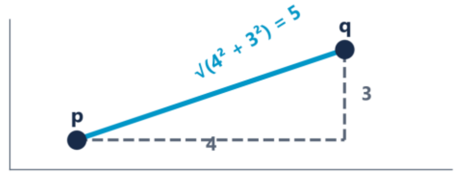
n the number of features (dimensions)

Σ sum over all n features

Squaring makes every difference positive AND penalises large gaps disproportionately - one big difference outweighs several small ones

Also called the L2 norm, or the 2-norm

Use it when features are continuous, on comparable scales, and straight-line closeness is meaningful



ON PAPER p = (2,3), q = (6,6). Compute it before I do.

Manhattan Distance - City Blocks

UNIT 1

You cannot walk through buildings - so sometimes the straight line is the wrong measure.

MANHATTAN DISTANCE IN 2 DIMENSIONS

$$d(p, q) = |p_1 - q_1| + |p_2 - q_2|$$

Sum of absolute differences. No squaring, no square root.

MANHATTAN DISTANCE IN N DIMENSIONS

$$d(p, q) = \Sigma_{i=1}^{n} |p_i - q_i|$$

Also called the L1 norm, taxicab distance, or city-block distance.

WORKED: THE SAME P = (2, 3), Q = (6, 6)

$$d = |2-6| + |3-6| = 4 + 3 = 7 \quad \text{(Euclidean was 5)}$$

Same two points. Different metric. Different answer. 7 vs 5.

WHERE, AND WHEN TO USE IT

| · | absolute value - drop the minus sign

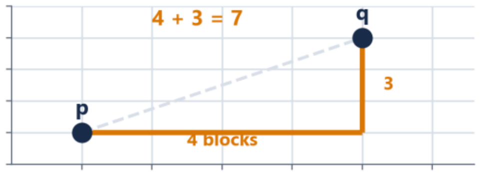
No squaring so every feature contributes in proportion to its difference - one big gap does NOT dominate

Also called L1 norm, taxicab distance, city-block distance

More robust to outliers than Euclidean, precisely because it does not square

Better in high dimensions where Euclidean distances tend to concentrate and lose contrast

Use it when movement is constrained to axes, or features are independent counts, or outliers are a concern



SAME TWO POINTS Why is the answer 7 and not 5?

Euclidean vs Manhattan - Same Points, Different Worlds

Neither is correct in general. Each is correct for a different question.

	Euclidean (L2)	Manhattan (L1)
Physical picture	As the crow flies - straight line	As the taxi drives - along the grid
Formula	$\sqrt{\Sigma(p_i - q_i)^2}$	$\Sigma\lvert p_i - q_i\rvert$
p=(2,3), q=(6,6)	5	7
Effect of one big gap	Dominates - squaring amplifies it	Contributes in proportion - no amplification
Outlier sensitivity	High	Lower - more robust
Shape of 'distance 1'	A circle	A diamond
Choose it when	Continuous, comparable features; straight-line closeness is meaningful	Axis-constrained movement; count features; outliers present; high dimensions

DECIDE A delivery robot moving along warehouse aisles. Which metric describes its real travel distance - and why?

Hands-On: Both Metrics on Three Customers

THE PROBLEM

- Customer A = (visits: 2, spend: 5)
- Customer B = (visits: 5, spend: 9)
- Customer C = (visits: 8, spend: 5)
- Question: which customer is A more similar to - B or C?

WHY IT MATTERS

- Euclidean punishes the single large gap (6 in visits) more heavily
- Manhattan treats a gap of 6 in one feature the same as 3+3 across two
- If a nearest-neighbour answer flips, so will your clusters

EUCLIDEAN	MANHATTAN
B	C
5.00 beats 6.00	6 beats 7

same three customers, opposite answers

STEP-BY-STEP SOLUTION

Euclidean A→B $\sqrt{[(2-5)^2 + (5-9)^2]} = \sqrt{[9 + 16]} = \sqrt{25} = 5.00$

Euclidean A→C $\sqrt{[(2-8)^2 + (5-5)^2]} = \sqrt{[36 + 0]} = \sqrt{36} = 6.00$

Euclidean verdict A is closer to B (5.00 < 6.00). So B is more similar.

Manhattan A→B $|2-5| + |5-9| = 3 + 4 = 7$

Manhattan A→C $|2-8| + |5-5| = 6 + 0 = 6$

Manhattan verdict A is closer to C (6 < 7). So C is more similar.

The point The two metrics DISAGREE about who A's nearest neighbour is. Same three customers, same numbers, opposite conclusions.

ANSWER **Euclidean says B. Manhattan says C. The metric changed the answer.**

1.4 DISTANCE AND SIMILARITY

Cosine Similarity - Direction, Not Magnitude

UNIT 1

Ignores how BIG the vectors are and asks only whether they point the same way.

COSINE SIMILARITY

$$\cos(\theta) = (A \cdot B) \,/\, (\|A\|\ \|B\|) = \Sigma\, A_iB_i \,/\, (\sqrt{\Sigma\, A_i^2} \cdot \sqrt{\Sigma\, B_i^2})$$

The cosine of the angle between two vectors. Range −1 to +1; for non-negative data, 0 to 1.

COSINE DISTANCE

$$d(A, B) = 1 - \cos(\theta)$$

Converts similarity to distance so it can be used where a distance is expected. 0 = identical direction.

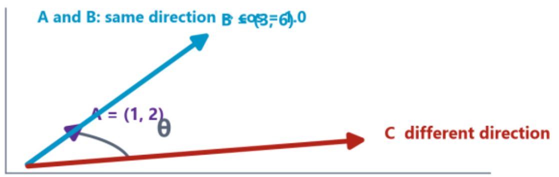
WORKED: A = (1, 2), B = (3, 6)

$$A{\cdot}B = 3 + 12 = 15 \;\cdot\; \|A\| = \sqrt{5} \;\cdot\; \|B\| = \sqrt{45} \;\rightarrow\; 15 \,/\, \sqrt{225} = 1.0$$

Perfectly similar - B is exactly 3 × A. Same direction, different magnitude.

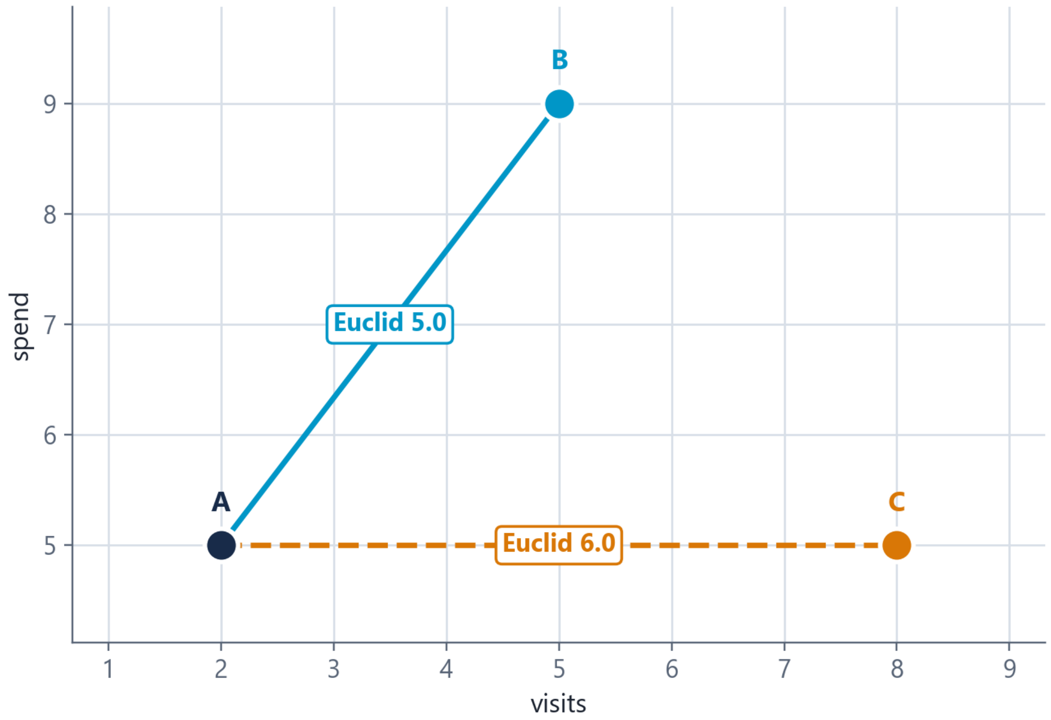
WHERE, AND WHEN TO USE IT

- $A \cdot B$ dot product - multiply matching components and sum them
- $\|A\|$ the length (magnitude) of vector A = $\sqrt{\Sigma\, A_i^2}$
- cos = 1 identical direction - as similar as possible
- cos = 0 perpendicular - completely unrelated
- cos = −1 opposite direction (impossible for count data, which is never negative)
- **Use it when** magnitude is irrelevant or misleading - text, ratings, sparse high-dimensional counts



PREDICT A 200-word and a 2,000-word report on the same match. Similar or not? Which metric agrees with you?

Euclidean and Manhattan Disagree - Drawn



Euclidean picks B (5.0 < 6.0). Manhattan picks C (6 < 7). Same three points.

HOW TO READ THIS FIGURE

A → B differs by 3 in visits AND 4 in spend - spread across both features.

A → C differs by 6 in visits and 0 in spend - one big gap, concentrated.

Euclidean squares so 36 beats 25. The single big gap is punished harder. B wins.

Manhattan adds so 6 beats 7. Spreading the difference costs more. C wins.

Same points opposite answers to 'who is A's nearest neighbour?'

TAKEAWAY If the metric can flip the nearest neighbour of three points, think what it does to 50,000.

Cosine vs Euclidean - Which Question Are You Asking?

Clustering by (Age, Income) is a Euclidean question. Clustering by purchased-item lists is a Cosine question.

Euclidean asks: how far apart?

Sensitive to magnitude. Two documents with the same proportions but different lengths are far apart.

Good for physical measurements, coordinates, standardised continuous features where size genuinely matters.

Example where it is right two customers with similar visit counts and similar spend really ARE similar customers.

Example where it fails a 200-word and a 2,000-word article on the same match look completely different.

Watch out for unscaled features - the largest-range feature silently takes over.

Cosine asks: same direction?

Sensitive to proportion only. Magnitude is divided out entirely.

Good for text, ratings, sparse high-dimensional counts - anywhere 'how much' is not the point.

Example where it is right recommending news articles on the same topic regardless of article length.

Example where it fails if magnitude genuinely matters. A customer spending Rs. 100 and one spending Rs. 100,000 in the same proportions look IDENTICAL to cosine.

Watch out for using it when 'how much' is exactly what the business cares about.

(Age, Income)
magnitude MEANS something
→ Euclidean

Ratings 1-10
some raters are generous
→ Cosine

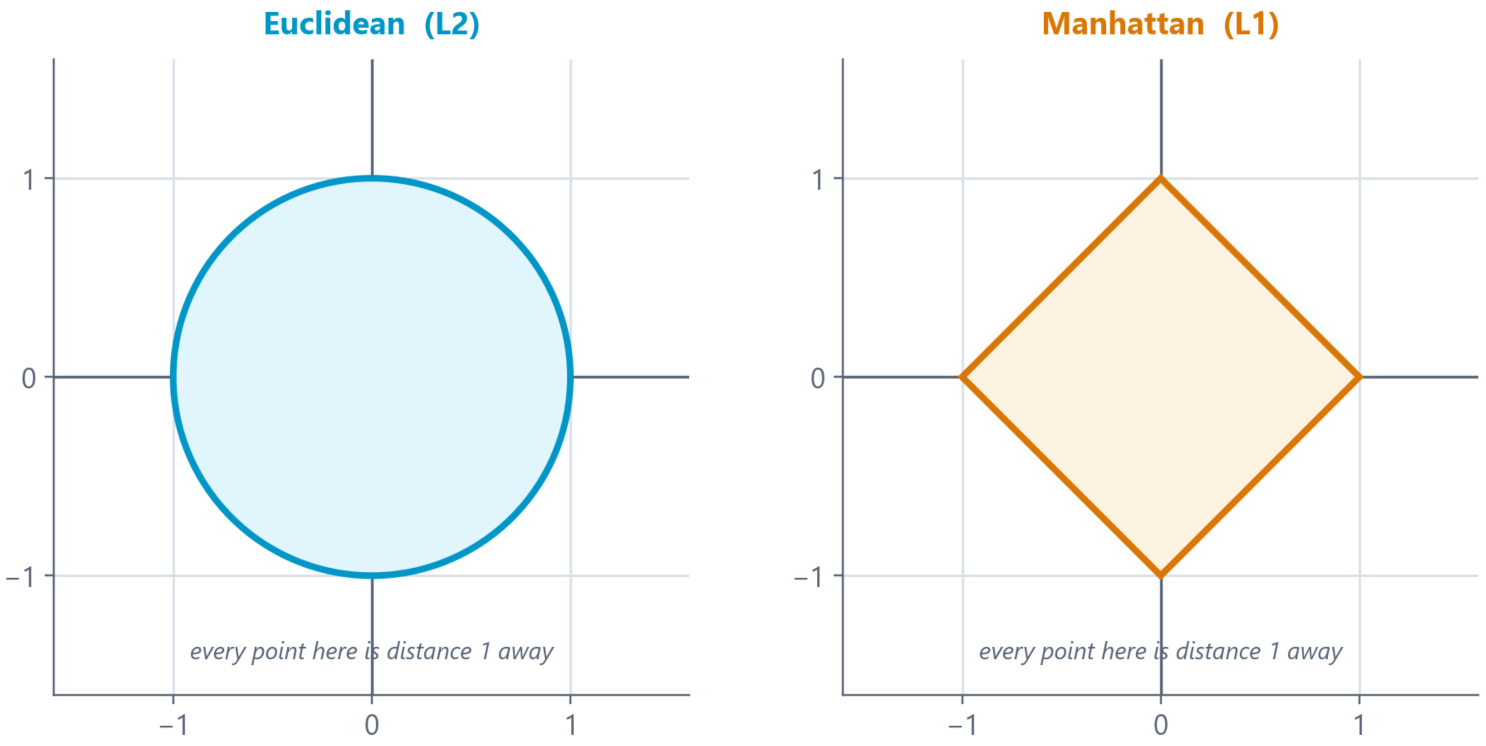
Purchased items
list length is noise
→ Cosine

Customer VALUE
spend size is the point
→ Euclidean

CHOOSE Recommending films from ratings out of 10, where some users rate everything high and some rate everything low. Which metric - and why?

1.4 DISTANCE AND SIMILARITY

'Nearby' Is a Different Shape in Each Metric



HOW TO READ THIS FIGURE

Both shapes contain every point at distance exactly 1 from the centre.

Euclidean gives a CIRCLE - the same in every direction.

Manhattan gives a DIAMOND - reaching further along the axes than diagonally.

Why the diamond moving diagonally costs you in BOTH coordinates, and Manhattan adds them.

The consequence clusters inherit these shapes - which is why the same algorithm gives different groups.

TAKEAWAY **Choosing a metric is choosing what shape 'nearby' has.**

05

SECTION

When and Why the Choice of Distance Matters

IN THIS SECTION

- The scaling pitfall - and how to fix it
- A worked demonstration: same data, opposite answers
- The curse of dimensionality
- A decision guide for choosing a metric

The Scaling Pitfall - The Most Common Error in This Subject

COMMON PITFALL · Unequal ranges mean unequal influence - silently

The setup you cluster people using two features: HEIGHT in centimetres (range roughly 150-190) and WEIGHT in kilograms (range roughly 45-100).

What you assume both features contribute equally to the distance. It feels fair - one feature each.

What actually happens height varies over ~40 units, weight over ~55. Both contribute, but not equally - and if one were in MILLIMETRES the imbalance becomes enormous.

The extreme version cluster customers by AGE (18-70) and INCOME (20,000-2,000,000). Income spans a range 30,000 times larger. Age contributes essentially nothing to the distance.

The consequence your 'customer segments' are income bands wearing a disguise. Age is in the model and has no influence whatsoever.

Why it is dangerous nothing errors. Nothing warns you. The code runs, the clusters look plausible, and the conclusion is wrong.

RANGE OF EACH FEATURE

squared: 2,700 vs 4,000,000,000,000

PREDICT Unscaled, what percentage of the distance does age contribute? Write a number.

The Fix: Put Every Feature on the Same Scale

UNIT 1

Standardise before you compute distances. This is not optional polish - it decides your answer.

STANDARDISATION (Z-SCORE) - THE USUAL DEFAULT

$$z = (x - \mu) / \sigma$$

Each feature ends with mean 0 and standard deviation 1. Keeps outliers visible. Use unless you have a reason not to.

MIN-MAX NORMALISATION

$$x' = (x - x_{min}) / (x_{max} - x_{min})$$

Squeezes every feature into [0, 1]. Bounded and intuitive - but one extreme outlier compresses everything else.

WORKED: INCOME 2,00,000 IN A SET WITH M = 5,00,000, Σ = 3,00,000

$$z = (200000 - 500000) / 300000 = -1.0$$

One standard deviation below the mean - now directly comparable with a z-score for age.

WHERE, AND WHEN TO USE IT

μ the mean of that feature across all objects

σ the standard deviation of that feature

After z-scoring every feature has mean 0, std 1 - so all features get equal say

Z-score vs min-max z-score keeps outliers visible; min-max lets one outlier squash everything else

Do NOT scale when features are already comparable (all counts of the same kind) or when relative magnitude IS the signal

Order matters fit the scaler, then compute distances. Never scale after clustering.

STANDARDISATION

$$z = (x - \mu) / \sigma$$

how many standard deviations from normal —

and that is DIMENSIONLESS

HANDS-ON DEMONSTRATION

Demonstration: Same Data, Unscaled vs Scaled

UNIT 1

THE PROBLEM

- Customer P = (age 25, income 3,00,000)
- Customer Q = (age 26, income 9,00,000)
- Customer R = (age 55, income 3,20,000)
- Question: which customer is P more similar to - Q or R?

WHY IT MATTERS

- Unscaled, a 30-year age gap counted for less than a 20,000 income gap
- After scaling, a 2σ difference in either feature carries equal weight
- Nothing in the code changed - only the units the data was expressed in

UNSCALED
R wins
by a factor of about 30

STANDARDISED
a tie
both ≈ 2.00

only the units changed

STEP-BY-STEP SOLUTION

UNSCALED, Euclidean P→Q $\sqrt{[(25-26)^2 + (300000-900000)^2]} = \sqrt{[1 + 3.6\times10^{11}]} \approx 600{,}000$

UNSCALED, Euclidean P→R $\sqrt{[(25-55)^2 + (300000-320000)^2]} = \sqrt{[900 + 4\times10^8]} \approx 20{,}000$

Unscaled verdict P is far closer to R. Age contributed 900 against hundreds of millions - it was invisible.

Now standardise suppose age has μ=40, σ=15 and income has μ=5,00,000, σ=3,00,000.

SCALED P = (−1.00, −0.67) SCALED Q = (−0.93, 1.33), SCALED R = (1.00, −0.60)

SCALED Euclidean P→Q $\sqrt{[(-1.00+0.93)^2 + (-0.67-1.33)^2]} = \sqrt{[0.005 + 4.00]} \approx 2.00$

SCALED Euclidean P→R $\sqrt{[(-1.00-1.00)^2 + (-0.67+0.60)^2]} = \sqrt{[4.00 + 0.005]} \approx 2.00$

Scaled verdict Now essentially TIED - because P differs from Q by 2σ of income, and from R by 2σ of age. Both features finally count.

ANSWER **Unscaled: R wins easily. Scaled: it is a tie. Scaling changed the answer.**

The Curse of Dimensionality - Distance Gets Less Useful as Features Grow

UNIT 1

COMMON PITFALL · In high dimensions, everything is roughly equally far apart

The phenomenon as you add features, the distance between the nearest and the farthest point shrinks towards zero RELATIVE to the distances themselves.

Why it happens every extra dimension adds another squared difference to the sum. With enough dimensions, every pair accumulates a similar total - so all distances converge.

The consequence 'nearest neighbour' stops being meaningful. If everything is equally far, clustering has nothing to grip on.

Symptoms you will see clusters that look arbitrary; results that change completely on a small data change; silhouette scores hovering near zero.

What to do about it select fewer, better features; reduce dimensions (PCA, t-SNE - your Unit III); or use cosine similarity, which is more robust in sparse high-dimensional spaces.

The counter-intuitive lesson adding a feature is not free. An irrelevant feature does not just fail to help - it actively dilutes every distance you compute.

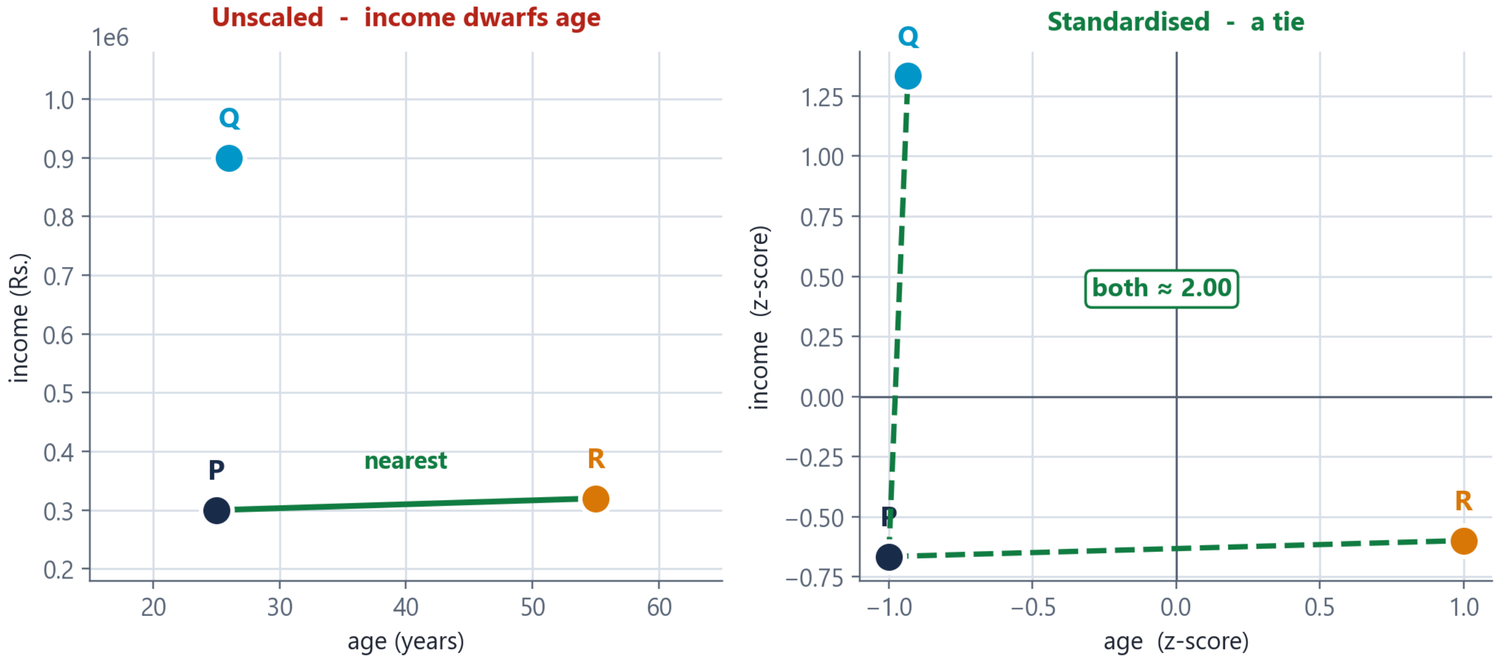
THREE SYMPTOMS IN THE LAB

1. clusters look arbitrary
2. results flip on a re-run
3. silhouette hovers near zero

→ suspect dimensions, not the algorithm

PREDICT You have 50 customers and 300 features. What do you expect your clustering to look like - and why?

The Same Three Customers, Unscaled and Scaled



HOW TO READ THIS FIGURE

Left - unscaled the axes are age (0-60) and income (up to 10 lakh). Age is invisible.

The green line P's nearest neighbour is R, by a factor of about 30.

Right - standardised both axes are now z-scores, so both span roughly −2 to +2.

The two dashed lines P→Q and P→R are now the same length. A dead tie.

Nothing changed except the units. Same data, same algorithm, reversed answer.

TAKEAWAY A 30-year age gap counted for less than a 20,000 income gap - until you standardised.

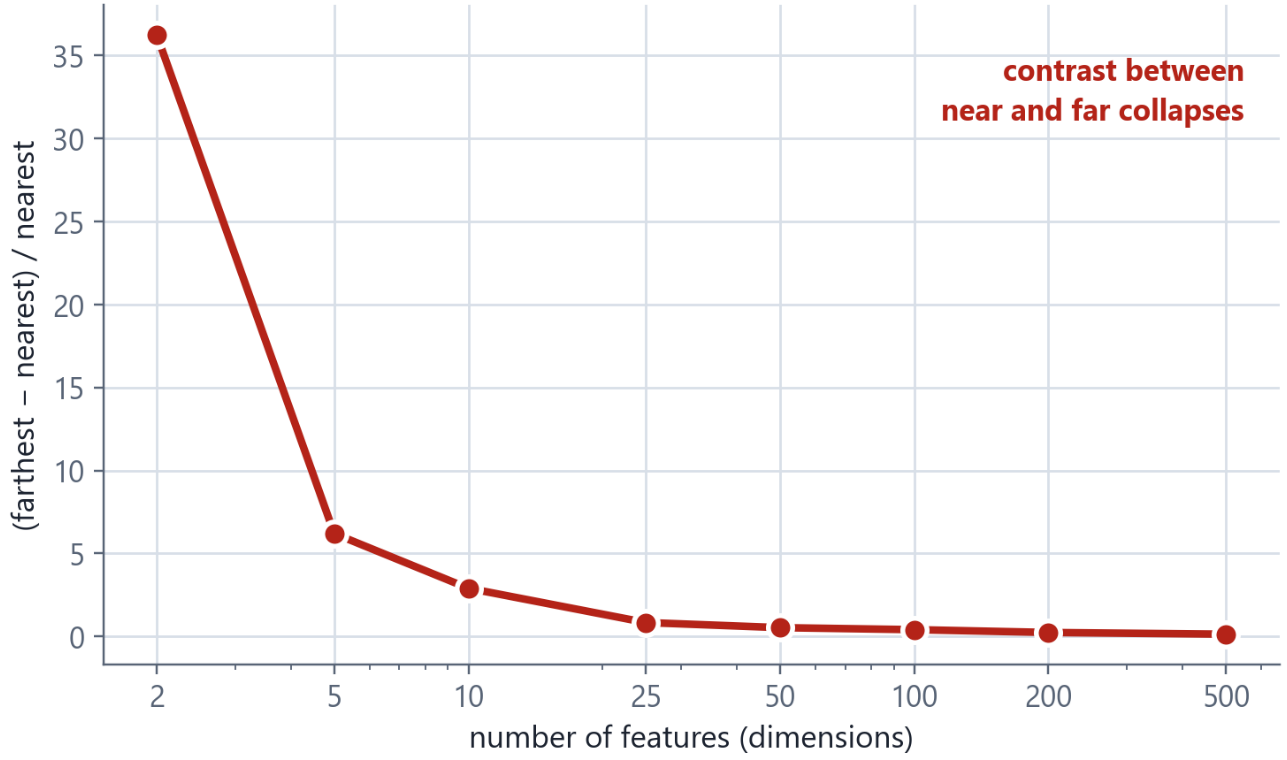
Choosing a Metric - A Decision Guide

Not a ranking - a mapping from the shape of your data to the right question to ask of it.

If your data is...	Use	Because
Continuous features on comparable scales (after standardising)	Euclidean	Straight-line closeness is meaningful and magnitude matters
Movement constrained to axes - grids, aisles, city blocks	Manhattan	It measures the distance actually travelled, not the impossible shortcut
Contains outliers you do not want dominating	Manhattan	No squaring, so one huge gap does not swamp the sum
Text, documents, or word counts	Cosine	Length is noise; the proportion of topics is the signal
Ratings where users differ in generosity	Cosine	Removes the overall level and keeps the pattern of preference
Sparse and very high-dimensional	Cosine	More robust than Euclidean when distances start to concentrate
Features on wildly different ranges	STANDARDISE FIRST	Then choose. No metric survives unscaled features.

USE IT NOW Take the dataset from your practical. Walk the table and pick a row.

The Curse of Dimensionality, Measured



HOW TO READ THIS FIGURE

The measurement pick a point; compare its nearest and its farthest neighbour.

Low dimensions the farthest is many times farther. Real contrast between near and far.

High dimensions the ratio collapses - everything is roughly equidistant.

Consequence 'nearest neighbour' stops meaning anything, so clustering has nothing to grip on.

Responses fewer, better features; dimensionality reduction; or cosine, which holds up better when data is sparse.

TAKEAWAY **Adding an irrelevant feature does not merely fail to help - it dilutes every distance you compute.**

CHECK YOUR UNDERSTANDING

Which Metric? - Rapid Fire

UNIT 1

QUICK CHECK

You are clustering research papers using word-frequency vectors. Papers range from 4-page letters to 60-page reviews. Which metric, and why?

A Euclidean, because word counts are continuous numbers

B Manhattan, because it is robust to the outlier long papers

C Cosine, because paper length is noise and topic proportion is the signal

D Euclidean after min-max normalisation of every word count

ANSWER: C A 60-page review has roughly 15× the word counts of a 4-page letter on the SAME topic - so Euclidean and Manhattan both see a huge gap where there is none. Cosine divides out magnitude entirely and compares only the direction of the vector, i.e. the proportion of each topic. This is the textbook case for cosine similarity.

THE ONE-LINE GUIDE

- **E** continuous, comparable, size matters
- **M** grid movement, counts, outliers
- **C** text, ratings, sparse - size is noise
- **!** different ranges? SCALE FIRST

Activity: Choose the Metric, Defend the Choice

GROUP WORK · 12 minutes

In groups of 3-4, take one scenario. Decide the features, whether to scale, and which metric - then defend it against the obvious alternative.

1

Pick a scenario

- Grouping LPU students by their canteen spending patterns
- Recommending songs from listening histories
- Detecting unusual electricity usage in campus buildings
- Grouping cities by climate measurements

2

Specify it

- What is the OBJECT - one row of your data?
- Which 3-5 features would you use?
- What are their likely ranges?

3

Decide and justify

- Do you need to standardise? Why or why not?
- Euclidean, Manhattan or Cosine?
- Finish the sentence: 'two objects are similar when ___'

4

Argue the other side

- Name the strongest alternative metric
- Say specifically what would go wrong if you used it
- If you genuinely cannot decide, say so and say why

DELIVERABLE One speaker per group: 60 seconds giving your object, your metric, your scaling decision, and the one sentence completing 'two objects are similar when ___'.

UNIT I RECAP

Unit I - The Eight Ideas Worth Remembering

UNIT 1

1 **Three paradigms, decided by which labels you have**
Supervised needs every label; unsupervised needs none; semi-supervised stretches a few across many. The paradigm is a property of the question, not of the data.

2 **Unsupervised learning has no ground truth**
So there is no accuracy score. Evaluate with internal measures, human judgement, or the downstream decision - and decide which BEFORE you run anything.

3 **Formulation is harder without labels, not easier**
The label silently defined the task, the success criterion, the relevant features and the stopping point. Remove it and all four become your decisions.

4 **Five specifications make a problem well-posed**
Objects · Features · Similarity · Structure sought · Success criterion. Anything left implicit gets decided by accident.

5 **The four use-case families share one shape**
The categories do not exist yet, labelling is impractical, data is abundant, discovery has value, and success is judged downstream.

6 **Three metrics, three questions**
Euclidean asks how far (crow flies). Manhattan asks how far along the grid (city blocks). Cosine asks whether they point the same way.

7 **Unscaled features silently destroy your analysis**
Age contributed 900 while income contributed 400 million. Standardise with $z = (x - \mu)/\sigma$ before computing any distance. Nothing will warn you.

8 **The metric IS the definition of 'similar'**
Same three customers, Euclidean said B and Manhattan said C. Change the metric and you change the answer - so choose it deliberately, not by default.

NEXT UP Unit II - Clustering Techniques: K-Means and choosing k, hierarchical clustering and dendrograms, DBSCAN for arbitrary shapes, and how to evaluate a clustering you cannot score.

Unit I Complete - Your Checklist

YOU SHOULD NOW BE ABLE TO

- Distinguish supervised, unsupervised and semi-supervised learning, with an example of each
- Explain why the paradigm is decided by the labels you have, not the algorithm you prefer
- State why reinforcement learning is not unsupervised learning
- Formulate an unsupervised problem using all five specifications
- Explain why there is no accuracy score, and give three ways to evaluate anyway
- Describe all four use-case families with a concrete example each
- Write down the Euclidean, Manhattan and cosine formulas from memory
- Compute all three by hand for a given pair of points
- State the four properties of a true metric
- Explain the difference between a similarity and a distance
- Explain the scaling pitfall and fix it with $z = (x - \mu)/\sigma$
- Explain the curse of dimensionality and name three responses to it
- Choose and DEFEND a metric for an unfamiliar dataset

PRACTICE & LAB WORK

- Compute all three metrics by hand for p = (1,4,2), q = (5,1,6)
- Redo the age/income example with your own numbers - watch the answer flip
- In Python: cluster the Iris data with and without StandardScaler, compare
- Find a dataset where cosine is clearly right and explain why in 3 sentences
- Install numpy, pandas, scikit-learn and matplotlib before the first lab
- Write the 'two objects are similar when ___' sentence for any three datasets

READ / EXPLORE

- Ankur A. Patel, Hands-On Unsupervised Learning Using Python - Chapter 1
- Vaibhav Verdhan, Supervised and Unsupervised Learning for Data Science (Manning) - foundations
- scikit-learn user guide: sklearn.metrics.pairwise - read the docstrings for euclidean_distances and cosine_similarity
- scikit-learn user guide: Preprocessing data - StandardScaler vs MinMaxScaler
- Explore: plot 20 random 2D points and draw the unit 'circle' under both L1 and L2 - see the diamond